What Are the Main Parts of a Computer and Their Functions?

Most of us use computers regularly, but do you know the parts of a computer and the components that make your PC actually work?

Computer hardware has evolved a lot since its beginnings. You can now create your own PC on spec depending on whether you want to edit videos or have a high-end gaming machine. Computers are made up of a lot of different parts and they all have their own unique role to play in making it function, whether you use your computer for designing software or just surfing the internet and watching silly videos.

But what is a computer really made of and how does it actually work? Here are some answers that will help you become much better educated on PC hardware.

Component	Function
Central Processing Unit (CPU)	Acts as the brain of the computer, performing calculations and executing instructions.
Memory (RAM)	Temporarily stores data and programs currently in use for quick access.
Storage (HDD/SSD)	Provides long-term storage for files, programs, and the operating system.
Motherboard	Connects and allows communication between all components of the computer.
Graphics Processing Unit (GPU)	Handles rendering of images, videos, and animations, especially in games or design.
Power Supply Unit (PSU)	Converts electrical power to the correct voltage for the computer's components.
Cooling System	Prevents components from overheating by dissipating heat.
Input Devices	Allows users to interact with the computer (e.g., keyboard, mouse).
Output Devices	Displays or outputs results of computer processes (e.g., monitor, speakers).
Network Interface	Connects the computer to the internet or other networks for communication.

What Is a Computer?

A computer is basically a **machine that executes operations and calculations**. It is, in fact, a very complex calculator.

Nowadays, a computer can be a PC or a Mac - though, all rivalry aside, they are both the same thing at their core: let's take a moment to remember that PC simply means Personal Computer. It is there for a private user. That user can make it perform complex operations and interact with it using various hardware and software interfaces. It first appeared at the beginning of the 1980s when IBM launched its first PC/XT.

A simple overview of what a computer actually is and what its role may be.

Unlike the other "micro-computers" of the time, these computers were made out of standard components that were easy to duplicate.

Today, the PC (whether from Apple or another firm such as Asus or Dell) is still the most common type of computer in use today. There are a myriad of other computer types, of course - for example, a car's on-board computer or the computer that powers industrial robots or the metro.

What Is a PC Composed of?

Whether it's a tower PC, a desktop unit or a laptop, all computers have the same components:

- **A screen or monitor**
- **A keyboard**
- **A mouse or touchpad/trackball for laptops**
- **A central unit - the heart and mind of the computer**

The Computer Monitor

The monitor is the part of the computer that displays the data. It is part of the human-machine interface. It is thanks to the computer screen that the user can interact with the computer. Typing on the keyboard or using a mouse modifies the data that is displayed accordingly.

Monitors can vary greatly in size. The history of computers is fascinating and as you would expect, monitors have evolved a lot – there were initially some rather "LoFi" monitors out there with very low resolution displays.

The numbers - usually given in inches - designate the screen's diagonal. There are very compact laptop screens: 10 inches for the smallest laptops and netbooks. Office computers start at around 15 inches and can go up to 27 inches or more, especially if you decide to hook the computer up to a TV screen.

Resolution and prices vary as well. For example, a 24 inch screen usually has a resolution of 1920 x 1080 pixels, or high definition. Screens can be plugged in using the VGA, DVi or HDMI connections.

Did you think the monitor was the actual PC? You might want to invest in IT classes - our Superprof tutors are eager to help you become computer-literate.

The Keyboard

One of the computer's external hardware components. It lets you write text and communicate with the computer. Keyboards generally have about a hundred keys: the **letters of the alphabet**, **numbers**, **punctuation marks** and **special functions**. It is, in a way, the descendant of the typewriter.

There are several types of keyboards: wireless or not, with touchpad or without, gaming keyboards, specialized keyboards with additional keys and functions, and many more!



English-speaking countries use keyboards where the top row of letters starts with QWERTY. Photo credit: Betchaboy

The setup of the keyboards varies according to the language: English-speaking countries use the QWERTY keyboard (the first letters of the top row); other countries have different setups.

The Computer Mouse



The mouse is what lets you move the cursor (that little arrow) on the monitor. It lets you position it over certain elements and select them by clicking on the left button ("left-click") or access certain functions by clicking on the right one ("right-click). Most have a little wheel in the middle that lets you scroll up and down in text documents and websites (some have a touchpad in the middle instead). Whether you are exploring Facebook and other social media or you are playing a game, the mouse is a key piece of kit.

The first mouse was made of wood, with two rotary wheels for the X and Y axes, and only one button. Photo credit: Erik Pitti on Visual hunt

Mice have a motion captor on their underside. Now it is usually optical or laser-based, but some old-timers may remember the ball. You can buy plug-in mice that connect to your computer with a wire and wireless mice for which you only need to plug in a little captor to a USB port and make sure it is charged or change its batteries from time to time.

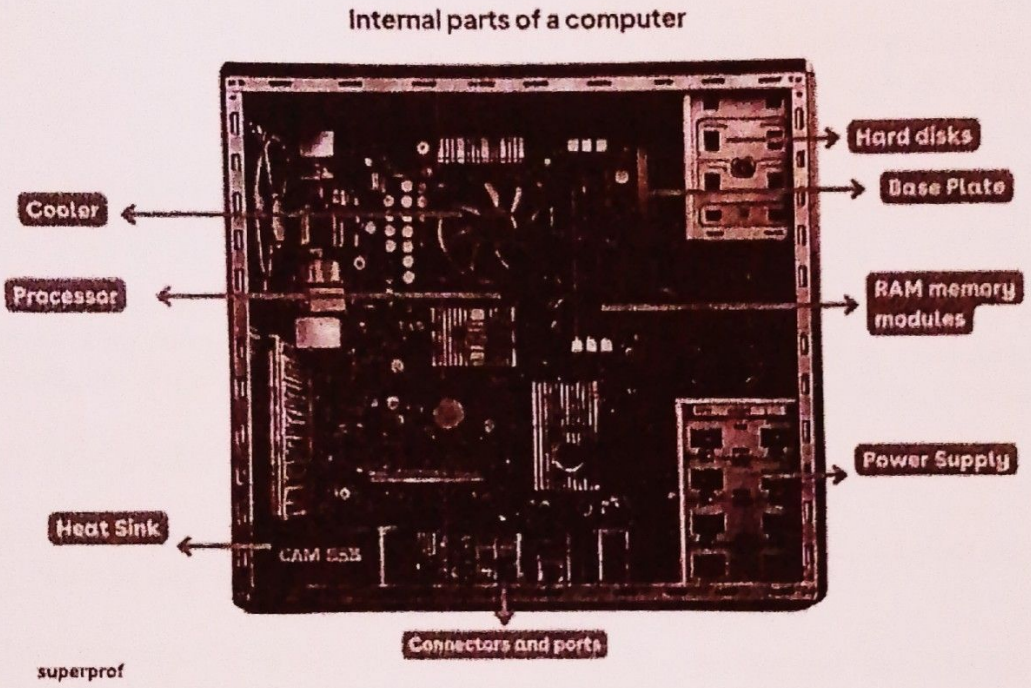
Learn about the surprisingly interesting history of the mouse!

A mousepad can help your mouse track correctly if the surface of your desk is a little rough or unusual. Laptops and netbooks may not need a mouse: they often have a trackball or touchpad

just under the keyboard that fulfills that role. You would indeed need a mouse or other pointing device if you were taking online computer courses!

The Computer's Central Unit

This is the casing that contains all the electronic material necessary for the computer to function. The keyboard, monitor and mouse are connected to it - with wires or via a wireless connection (Bluetooth, for example). This is also where you will attach USB devices and where your CD-ROM/BluRay player is located (assuming you still have something so old-fashioned). The internal parts of a computer are shown below, but there are some different variations of this depending on the setup.



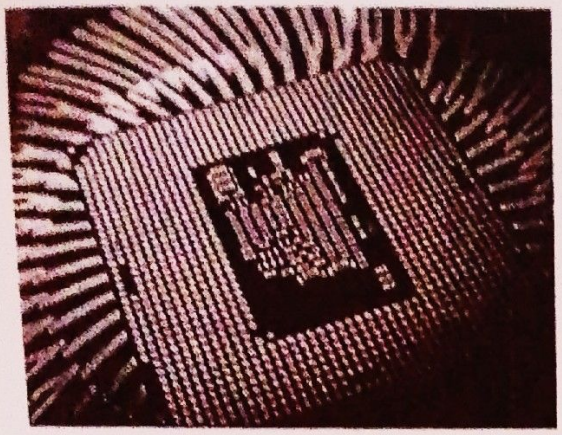
Now, some computers group everything together, and the central unit can sometimes be integrated into the monitor (for iMacs, for example) or under the keyboard (laptops).

The central unit will contain the CPU (the brain), the hard drive (memory), the motherboard (the central nervous system) and the power unit (heart and lungs).

Anatomy of a Central Unit

PC components are not visible from the outside - you have to open the casing to see what electronics make it work. Underneath it you will find a CPU as well as a hard drive, motherboard, power unit, RAM, and a graphics card.

The CPU



The **Central Processing Unit** or CPU is the computer's brain. It organizes the data exchange between the various components (RAM, hard drive, graphic card…)

Its main tasks are:

- Accessing data stored in memory
- Processing data
- Storing data in the memory

It is the CPU that does the calculations that allow the user to interact with the computer and display data on the monitor. At the time of writing, processors can reach 3Ghz and some computers have several processors working in tandem.

An Intel processor for a computer CPU. Photo credit: stratman² (2 many pix!) on Visualhunt.com / CC BY-NC-ND

This means they are capable of **processing billions of information packets per second** and accomplishing complex calculations that are at this moment pioneering advances in science and medicine. Your computing power will depend on the type of processor you have.

Your Computer's Hard Drive

While it's important to consider what kind of processor is integrated into your computer, don't forget to look at its hard drive, too.

It's your computer's long-term memory. Its role is to save computer data. The hard drive containes the operating system, whether it's Windows, MacOS or Linux, the programmes you installed, your files and your other data.

Hard Drives Keep Growing

The average capacity of new hard disk drives has grown from 1TB in 2015 to over 10TB in 2024.

It stocks information using a binary system. Today, you can save several terabytes of information (over 1 000 gigabytes), the equivalent of hundreds of thousands of photographs, thousands of videos or millions of text documents. As the world's best and most famous programmers have extended what computers can do, the demand for HDD space has increased.

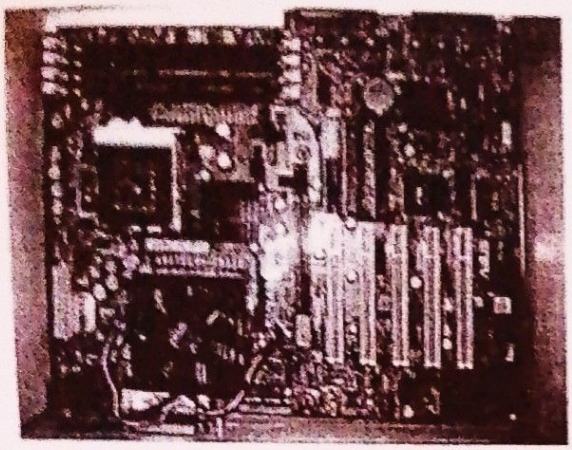
There are several types of hard drive:

Classical hard drives: They contain mechanical pieces including a read head that positions magnetic discs to read and write data

SSD hard drives: They do no have mechanical pieces and can retrieve data more rapidly.

External hard drives: Used to backup your data and your internal hard drive.

The Motherboard



The motherboard is the main component of the central unit. It centralizes and processes the data exchanged within the computer with the help of the processor, which is attached to it. It links the hard drive, mouse, keyboard, network, USB ports, etc. together.

A motherboard with its attachments. Photo credit: Iwan Gabovitch on Visual hunt / CC BY

This is the central nervous system onto which all the other computer components are plugged in.

The motherboard is a printed circuit connected to a chipset (a set of components that control almost of all of the motherboard). The chipset components are soldered directly onto the motherboard, and it is they that dictate what sort of processors and memory chips can be used with this computer. From 1995 on, almost all motherboards use the ATX format.

The Computer's Power Supply Unit

Obviously, nothing works without electricity. You computer receives its power through the power supply. It transforms the power from the wall socket to the right voltage for the computer and sends it on to the motherboard. It is also directly linked to certain mechanical components such as the CD-ROM drive.

Transforming the current results in a loss of energy that takes the form of heat. Therefore, a ventilation system is also installed in an adjacent unit and expulses the warm air through the back of the computer. 400 Watts is usually enough, but some power supply units can go up to 1000 watts!

Working Memory, or RAM

RAM (Random Access Memory) is a type of volatile memory accessed by the processor. It stores the data currently in use.

The RAM's particularities are:

- Its fast access speed
- Its temporary aspect: all data in the RAM is lost once the computer is turned off.

RAM memory cards range from 256 megabytes to 2 gigabytes depending on the processor, the motherboard's computing capacities and the use you make of your computer.

You would go much more in depth on the study of these components in your IT courses.

The Graphic Card

If you are a gamer, a good graphic card is important. It takes over the conversion of computer code into visual data, taking some of the workload from the processor, which can use the free

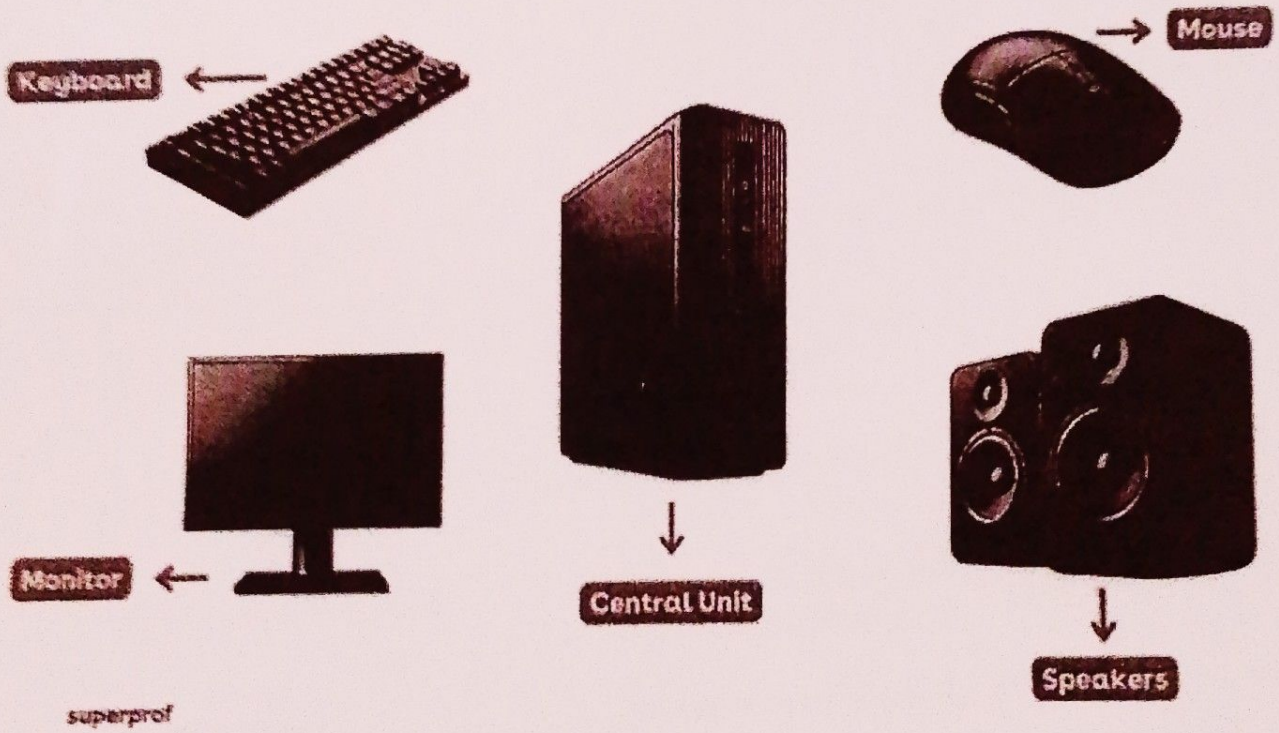
capacity to load other game data. A graphic card is useful as soon as you are working with high-quality images - as a graphic designer, for example, or a printer. Alternatively, the motherboard chipset can also integrate a graphic card.

Additional Computer Parts

These days, most screens include a webcam. But there are other accessories including external webcams that you can attach to your computer to expand its functions:

- A printer
- A scanner
- A webcam (usually integrated on the screen)
- An internet connexion (by wire or WiFi)
- USB (Universal Serial Bus) sticks or other devices
- A memory card
- A controller

External parts of a computer



Today, almost all external hardware can be attached through the USB port - no more guessing which port to use for your mouse or printer!

With Superprof, you can learn more about computers and how they work with one of many IT tutors that we can connect you to. Whether you are a total beginner or you are looking to work in computing, you can enhance your knowledge in a one-to-one setting with Superprof.

What are the Most Important Concepts in Fundamentals of Computer Science?

What makes a machine intelligent? How does your favourite app respond in milliseconds? What's behind the scenes of the technology you rely on every day?

The answers to these questions lie in the fundamentals of computer science.

Whether you're developing a mobile app, managing cloud infrastructure or creating algorithms for smart devices, a deep understanding of computer science fundamentals is essential. These concepts are the building blocks of digital systems, and they underpin everything from social platforms to cybersecurity networks.

In this blog post, we'll unpack the basics, explore the applications of computer science and spotlight the latest trends in computer science driving tomorrow's breakthroughs. This is your comprehensive guide to computer science basics for beginners, and how to launch your career in the field.

What is Computer Science?

Computer science is the study of computers and computational systems. Fundamental concepts in computer science span everything from the theoretical underpinnings of algorithms to the hands-on creation of new software.

At its heart, computer science is about solving problems – harnessing technology to tackle real-world challenges. From hardware design to data security, computer science covers a wide range of disciplines that combine creativity, critical thinking and technical knowledge.

Importance and Applications of Computer Science

So why is computer science relevant to you? Because almost every sector needs digital transformation.

Key applications of computer science include:

- **Healthcare**: Developing diagnostic tools and managing patient data.
- **Finance**: Automating trading systems and enhancing cybersecurity.

- **Education**: Creating adaptive learning platforms and virtual classrooms.
- **Entertainment**: Powering video games, streaming services, and virtual reality experiences.

The importance of computer science, sitting at the heart of so many important industries, means more opportunities for those who understand it. If you know how to design systems, protect data and build intelligent software, the career possibilities are endless.

Core Concepts in Fundamentals of Computer Science Every Student Should Know

You don't need to know everything on day one. But if you want to build a strong foundation, here's where to start.

1. Algorithms and Their Importance in Problem-Solving

Algorithms are essentially instructions. They define how a computer processes information to solve problems. A good algorithm saves time, memory and processing power. You'll use them in almost every area of software development – from sorting data to powering AI.

2. Data Structures: Types & Uses

Want fast access to data? You'll need to choose the right structure. Arrays, stacks, queues, trees and graphs – each has its purpose. Some are better for searching. Others are ideal for storing relationships between elements. When you understand how data is structured, you unlock the potential to build smarter, faster systems.

3. Programming Languages and Paradigms

Think of programming languages as the tools you use to communicate with a computer. Python, Java, JavaScript and C++ are just a few. They all offer different strengths and levels of complexity.

Importance of Understanding Programming Languages

Some languages are better for rapid prototyping, whereas others give you more control over hardware. Knowing what to use, and when, can make your development process more efficient.

Basic Programming Concepts

Fundamental programming concepts include variables, control structures (loops and conditionals), functions and error handling. These basics are the building blocks for writing clear and effective code.

Types of Paradigms

- **Procedural**: Step-by-step instructions, often used for simple tasks.
- **Object-Oriented**: Data and functions are grouped into objects, great for managing complexity.
- **Functional**: Emphasises immutability and reusable functions.
- **Logic-Based**: Solves problems by defining rules and relationships.

Learning multiple paradigms gives you flexibility in how you approach problems, which is a key advantage in real-world development.

4. Computer Architecture and Hardware Basics

Understanding what's inside a computer helps you write better software. Learn how CPUs, memory, input/output devices and storage work together. If you know how data moves through hardware, you can make your code more efficient.

5. Computer Networking and Internet Fundamentals

Every time you send a message, stream a video or visit a website, networking protocols are in play. Learn how IP addresses, DNS, routing and encryption work to keep systems connected and secure.

6. Databases and Data Management

Data powers everything, but only if it's well organised. Learn the basics of relational and non-relational databases, and understand how to query them using SQL. Good database design can mean the difference between a lightning-fast app and one that frustrates users.

7. Software Development Life Cycle (SDLC)

Software isn't built in one step. It evolves from planning and design to testing, deployment and maintenance. The SDLC gives you a framework to manage this process, reduce risk and improve quality.

Emerging Trends and Technologies in Computer Science

What's changing? A lot. And fast.

Here's a glimpse into the latest trends in computer science.

1. Artificial Intelligence (AI) and Machine Learning (ML)

AI and ML are already changing how we work, shop, communicate and make decisions. From chatbots to recommendation engines, these technologies require strong foundations in statistics, data science and programming.

2. Blockchain and Its Applications

When most people hear the word "blockchain", they just think of crypto. But its applications are far wider than that, touching on themes of trust, transparency and decentralisation. Use cases now include identity verification, supply chain management and smart contracts.

3. Cloud Computing and Edge Computing

Cloud platforms like AWS, Azure and Google Cloud make computing power accessible to everyone. Meanwhile, edge computing brings processing closer to the user, reducing latency for real-time applications like autonomous vehicles or IoT devices.

4. The Importance of Data Science

At the intersection of computing and business strategy, data science enables informed decision-making. By combining programming skills with statistical analysis, you can turn raw data into real value.

5. Cybersecurity and Its Importance

As systems grow more complex, so do threats. Cybersecurity is now a fundamental part of development, not an afterthought. Understanding encryption, secure coding practices and threat detection is essential for any tech professional.

Build a Successful Career in Computer Science with BSBI

At BSBI, our **BSc (Hons) Computer Science and Digitisation** programme is designed to provide you with a comprehensive understanding of both theoretical and practical computer science concepts. You'll gain hands-on experience, learn from industry experts and develop skills that are highly sought after in the job market.

Basic Counting and Numbers/ Exercises

- There are **seven** days in one week.
- "5" is an **odd** number.
- "6" is an **even** number.
- This is a **circle**.
- This is a **square**.
- The number **ten** is bigger than the number **five**.

1. Addition Sentences (Putting Together)

- **1 + 1 = 2** (One plus one equals two)
- **2 + 3 = 5** (Two **plus** three is five)
- **4 + 1 = 5** (Four and one add up to five)
- **7 + 5 = 12** (Seven plus five is equal to twelve)
- **8 + 2 = 10** :.....
- **10 + 10 = 20** :.....
- Addition is a commutative operation **(The order of addends does not change the sum)**

  E.g.: 4+5 = 5+4 = 9 (4 plus 5 equals 5+ 4 = 9)

- Adding zero to any number does not change the number

  E.g.: 3+0=3 (3 plus 0 equals 3)

2. Subtraction Sentences (Taking Away)

- **3 - 1 = 2** (Three **minus** one equals two)
- **5 - 2 = 3** (Five **take away** two is three)
- **10 - 4 = 6** (Ten minus four is six)
- **44 - 10 = 34** (Forty-four minus ten equals thirty-four)
- **9 - ? = 3** (Nine minus what equals three?).....

3. Multiplication Sentences (Groups Of)

- $2 \times 2 = 4$ (Two times two equals four)
- $3 \times 4 = 12$ (Three multiplied by four is twelve)
- $5 \times 4 = 20$ (Five times four is twenty)
- $5 \times 2 = 10$ (Five **groups of** two makes ten)

Example: If you have 3 plates with 2 oranges on each, you have

$3 \times 2 = 6$ oranges (**3 cross 2 equals 6**)

- **A positive number multiplied by a negative number** is negative.
- A **negative** number multiplied by a **negative** number is **positive**.

4. Division Sentences (Sharing Equally)

- $4 \div 2 = 2$ (Four divided by two is two)
- $10 \div 2 = 5$ (Ten divided by two equals five)
- $12 \div 3 = 4$ (Twelve divided by three is four)
- $35 \div 7 = 5$ (Thirty-five divided by seven is five)

- Division is sharing a collection of items into equal parts.
- The number being divided is the dividend.
- The number that divides the dividend is the divisor.
- The answer to a division problem is the quotient.
- Example: If you have 12 cookies to share among 3 friends, each friend gets

  $12 \div 3 = 4$ (12 divided by 3 equals 4)cookies.

- Estimate: To make a guess about a number or quantity.
- Even number: A number that can be divided by 2 without a remainder (e.g., 4, 6, 10).
- Odd number: A number that leaves a remainder of 1 when divided by 2 (e.g., 1, 3, 5, 9).

5. Inequality Sentences (Comparing Values)

- $5 > 3$ (Five is greater than three)

- $4 < 10$ (Four is less than ten)
- $10 + 5 > 11$ (The sum of ten and five is greater than eleven)
- $15 + 2 < 25$ (The sum of fifteen and two is less than twenty-five)

6. Sentences with unknowns (Variables)

- $4 + ? = 7$ (Four plus something equals seven)
- $x - 24 = 10$ (A number minus twenty-four equals ten)
- $3 \times y = 15$ (Three times a number equals fifteen)

Généralités

Saying /writing the date in English

1) the days of the week

En anglais...	En français...
Monday	lundi
Tuesday	mardi
Wednesday	mercredi (attention à la prononciation)
Thursday	jeudi
Friday	vendredi
Saturday	samedi
Sunday	dimanche

2) the months of the year

En anglais...	En français...
January	janvier
February	février
March	mars
April	avril
May	mai
June	juin

July	juillet
August	août
September	septembre
October	octobre
November	novembre
December	décembre

Pour les jours et les mois en anglais, je n'oublie pas la majuscule, alors qu'elle n'est pas nécessaire en français.

3) ordinal and cardinal numbers

On se souvient: un nombre ordinal, c'est un nombre qui permet de classer quelque chose dans un ordre: le 1er à arriver, le 3e à rendre sa copie, la 4e au niveau des résultats scolaires etc.

Généralement, on forme le nombre ordinal à partir du nombre cardinal, auquel on ajoute '-th' Exemple : 7 = seven => 7e = seventh

Malheureusement, il y a beaucoup d'exceptions, à commencer par les 3 premiers. Je vous ai signalé tous les nombres bizarres par

Vous voyez, il y en a beaucoup!

1	one	premier	first	1st
2	two	2e	second	2nd
3	three	3e	third	3rd
4	four	4e	fourth	4th
5	five	5e	fifth	5th
6	six	6e	sixth	6th
7	seven	7e	seventh	7th

8	eight	8e	eighth	8th
9	nine	9e	ninth	9th
10	ten	10e	tenth	10th
11	eleven	11e	eleventh	11th
12	twelve	12e	twelfth	12th
13	thirteen	13e	thirteenth	13th
14	fourteen	14e	fourteenth	14th
15	fifteen	15e	fifteenth	15th
16	sixteen	16e	sixteenth	16th
17	seventeen	17e	seventeenth	17th
18	eighteen	18e	eighteenth	18th
19	nineteen	19e	nineteenth	19th
20	twenty	20e	twentieth	20th

DE 21 A 100

A partir de 21, tout est logique, si on a assimilé le tableau précédent!

21	twenty-one	21st	twenty-first
22	twenty-two	22nd	twenty-second
23	twenty-three	23rd	twenty-third
24	twenty-four	24th	twenty-fourth
25	twenty-five	25th	twenty-fifth
26	twenty-six	26th	twenty-sixth
27	twenty-seven	27th	twenty-seventh
28	twenty-eight	28th	twenty-eighth
29	twenty-nine	29th	twenty-ninth

Bon, on n'en a pas vraiment besoin pour la date, mais autant tout apprendre aujourd'hui. Intéressons-nous aux dizaines intermédiaires qui vont vous permettre de former tous les nombres jusqu'à 100 :

30	thirty	30th	thirtieth
40	forty	40th	fortieth
50	fifty	50th	fiftieth
60	sixty	60th	sixtieth
70	seventy	70th	seventieth
80	eighty	80th	eightieth

90	ninety	90th	ninetieth
100	hundred	100th	hundredth

4) Ca y est! Formons la date.

La question qu'on pose pour demander la date:
- What is the day today?
- What's the date today?

ATTENTION !

Le format de la date dépend en fait du pays où vous êtes : il y a un système britannique et un système américain.

A l'oral, on ne vous embêtera pas et on vous comprendra, mais à l'écrit cela peut peut-être parfois poser problème.

En français : Jeudi 4 avril 2030

En ANGLAIS BRITANNIQUE :

Thursday 4th April 2030 OU Thursday 4 April 2030

En ANGLAIS AMERICAIN:

Thursday, April 4 2030 OU Thursday, April 4th 2030

A l'oral, on lit généralement : Thursday, the fourth of April two thousand thirty.

Littéralement : jeudi, le 4e jour d'avril 2030. Vous voyez, c'est logique, même si cela oblige les francophones à faire une certaine gymnastique : le jour, puis le rang, puis le mois et enfin l'année.

Comme vous l'avez compris, on dit par exemple à l'oral : "le 3e jour de mai".

C'est pour cela que vous allez avoir besoin du nombre ordinal (the first, the second, the third..), mais, à l'écrit, vous allez l'abréger par les deux dernières lettres.

Monday, May 1st
(the first)

Tuesday, May 2nd
(the second)

Wednesday, May, 3rd
(the third)

Thursday, May 4th
(the fourth)

... -th pour les suivants, y compris 11th (the eleventh), 12th (the twelfth), 13th (the thirteenth)...

SAUF 21st, 22nd, 23rd et 31st

Note : pour lire l'année à l'oral...
2003, 1997... Ces dates n'utilisent pas le système des nombres ordinaires.

La règle la plus courante: généralement, on coupe la date en 2 éléments:
19 | 97 = nineteen ninety-seven

SAUF à partir de l'an 2000, à partir duquel on lit la date comme un nombre ordinaire:

ANGLAIS BRITANNIQUE: 2001 = two thousand AND one

ANGLAIS AMERICAIN: 2001 = two thousand one

A compilation by Dr. Oladjidé Moutinou Soumanou (April2018)